JRG Welding

Structural & Ornamental Welding · Aluminum · Steel · Stainless

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WORK PROPOSAL

Proposal No.: **JRG-2026- [###]**

Date: **July 1, 2026**

Valid for: **30 days**

PREPARED FOR

Client: **[Client Name]**

Project Address: **18398 131st Trail N, Jupiter Farms, FL**

Phone / Email: **[Client Contact]**

PROJECT — TWO PARTS

A) Replacement of two (2) V-configuration wood roof rafter sections with structural aluminum members.

B) Replacement of the termite-damaged floor structure (2 main beams + 7 joists) with structural aluminum members. Like-for-like appearance and layout, equal or greater strength.

1. EXISTING CONDITIONS

A — Roof V-Sections

The residence is an A-frame type structure with exposed sloped roof rafters ("V" members) supporting the roof deck at the gable overhangs. The members scheduled for replacement are solid wood timbers with an actual cross-section of **3½" × 5½" (nominal 4×6)**. Two (2) V-sections are to be removed and replaced. Each V-section consists of two sloped members approximately **20'-0" long** meeting at the ridge, cantilevering approximately **1'-0" past the support beams** to form the eave overhangs. The members carry wood plank decking with a metal roof panel above, both of which will be re-fastened to the new aluminum members.

B — Floor Structure

The floor framing shows **severe, active termite damage** concentrated on one side of the system. The members to be replaced are: the two (2) parallel main beams ("girders") of **3" × 11½" × 13'-0"** running side by side, and the seven (7) floor joists of **2" × 11½" × 12'-0"** on the damaged side, hung from sheet-metal hangers. **The joists on the opposite side are in good condition and remain as-is**, and the existing wood beam at the far end of the damaged bay also remains. The sound continuation of the girder line beyond the columns likewise remains in wood: the new aluminum girders **meet the existing wood beams at the columns — each member bolted directly to the column**. Aluminum replacement permanently eliminates the termite and rot problem in the replaced members.

2. SCOPE OF WORK — PART A: ROOF V-SECTIONS

1. **Mobilization & protection:** Set up ladders/scaffolding as required; protect roofing, glazing, decking and landscaping in the work area.
2. **Temporary support:** Install temporary shoring to safely carry the roof load while each V-section is out.
3. **Demolition — protecting the roof at all times:** Carefully detach the wood plank decking and metal roof panel from the members being replaced, supporting and protecting the roof so nothing shifts or is damaged; cut free and remove the two (2) existing wood V-rafter sections in controlled segments (four sloped members total, approx. 20'-0" each). Haul away and legally dispose of debris.
4. **Materials:** Furnish new **6061-T6 structural aluminum rectangular tube** supplied by **Eastern Metal Supply** (size per Section 4; each 20' member from a single full-length extrusion — no mid-span splices).

5. **Fabrication:** Miter-cut members at the field-verified ridge angle. Fabricate two (2) concealed internal chevron-plate inserts: 3/8" 6061-T6 plates waterjet/plasma-cut in one piece to the exact ridge angle, joined with welded spacers, pre-drilled and tapped — each insert slides into both tubes as a single unit (see DWG-3). Welds per AWS D1.2.
6. **Finish:** Finish aluminum in **[Dark Bronze powder coat / wood-grain finish / paint to match]** to visually match the existing dark-stained timbers.
7. **Installation:** Set new aluminum members in the exact position of the removed timbers. Ridge joint assembled with 1/4"-20 stainless flat-head countersunk screws, flush and concealed. **Members fully sealed: welded/snap-in aluminum end caps at the eave tips, sealed watertight,** with 1/4" weep holes at low points. Isolate aluminum from treated wood and dissimilar metals with barrier tape/washers to prevent galvanic corrosion. Seal miter joint with wood-tone sealant.
8. **Roof deck & metal panel:** Re-install wood plank decking over the new members with #14 winged self-drilling wood-to-metal screws (410 SS or ceramic-coated, concealed under roofing), and metal roof panel with #10 woodgrip screws with EPDM washers; drip edge at eaves and ridge cap at the peak (see DWG-4).
9. **Closeout:** Remove shoring, touch-up finish, final cleanup and walkthrough with owner.

3. SCOPE OF WORK — PART B: FLOOR STRUCTURE REPLACEMENT

1. **Temporary shoring (critical):** Install temporary shoring posts and beams under the floor structure on both sides of the work area to fully carry the floor above while the beams and joists are out — nothing settles, nothing moves. Shoring remains in place until every new member is installed and fastened.
2. **Demolition — main beams:** Carefully cut free and remove the two (2) termite-damaged 3×12 main beams (13'-0" each), working in controlled sections and protecting the finished floor above and all adjacent framing.
3. **Demolition — joists (damaged side only):** Remove the seven (7) termite-damaged 2×12 floor joists (12'-0" each) on the affected side together with the old sheet-metal joist hangers. **The joists on the opposite side are sound and are not disturbed.** Haul away and legally dispose of all infested debris.
4. **Materials:** Furnish new 6061-T6 aluminum from Eastern Metal Supply: two (2) girders RT 3"×12"×3/16" cut to length, seven (7) joists RT 2"×12"×1/8" (joists cut two per 24' bar — no waste), aluminum angle for seats, and stainless hardware.
5. **Fabrication:** Weld and drill aluminum angle seats and top clips (3"×3"×1/4") for every joist-to-girder connection — far stronger than the sheet-metal hangers being removed, and nothing for termites to eat. Welds per AWS D1.2.
6. **Installation:** Set the new aluminum girders bearing on the existing wood columns. Where the sound wood beam continues past the column, **both members bolt directly to the column — no plates**: (2) 1/2" stainless thru-bolts pass through the aluminum girder tubes and the column, and (2) more pass through the wood beams and the column; isolation membrane at every aluminum-to-wood contact, tube ends sealed (see DWG-5). Condition of each post is verified during shoring; if a post is termite-damaged, its replacement is quoted separately before proceeding. Set each new aluminum joist **in the exact location of the existing one it replaces — spacing and layout match the existing floor exactly**: one end seated on the new aluminum girders, the other end seated on the **existing wood beam (which remains)** with aluminum seats, stainless lag screws and isolation barrier. The sound wood joists on the opposite side are re-supported on the new aluminum girder with new aluminum angle seats. Shim to existing floor elevation; isolation barriers at all wood contact points.
7. **Re-fasten the floor:** Re-screw all existing floor planks to the new aluminum joists with #14 winged self-drilling wood-to-metal screws (410 SS / ceramic-coated), pulling the floor back tight and flat.
8. **Closeout:** Remove shoring, final cleanup, and walkthrough with owner.

4. MATERIAL SELECTION — STRENGTH MATCH

There is no aluminum tube produced in the exact 3½" × 5½" timber dimension, so the closest standard extrusions stocked by Eastern Metal Supply were evaluated. The table compares each candidate against the existing wood member (Southern Pine No. 2, F_b = 1,000 psi, E = 1.4×10^6 psi; allowable aluminum bending stress 19.5 ksi per Aluminum Design Manual, E = 10.1×10^6 psi):

Member	Weight (lb/ft)	Stiffness EI (lb-in²)	Stiffness vs. wood	Bending capacity vs. wood
Existing wood 4×6 (3.5"×5.5")	4.8	6.79×10^7	1.00×	1.00×
Aluminum 3"×5"×1/8" 6061-T6	2.3	6.76×10^7	0.99×	3.0×
Aluminum 3"×5"×3/16" 6061-T6 SELECTED	**3.4**	**9.71×10^7**	**1.43×**	**4.2×**
Aluminum 4"×6"×3/16" 6061-T6 (heavier alternate)	4.2	1.84×10^8	2.7×	6.7×

Selected member: 3" × 5" × 3/16" wall rectangular tube, alloy 6061-T6 (Eastern Metal Supply). It keeps a slim profile close to the existing timber (½" smaller in each direction), provides **1.4× the stiffness and over 4× the bending strength** of the wood member it replaces, welds cleanly at 3/16" wall, and weighs 30% less than the timber. The 3/16" wall was chosen over 1/8" because it keeps the assembly stiffer than the original member with comfortable margin even if the existing lumber were a stiffer species (e.g. Douglas Fir).

B — Floor members

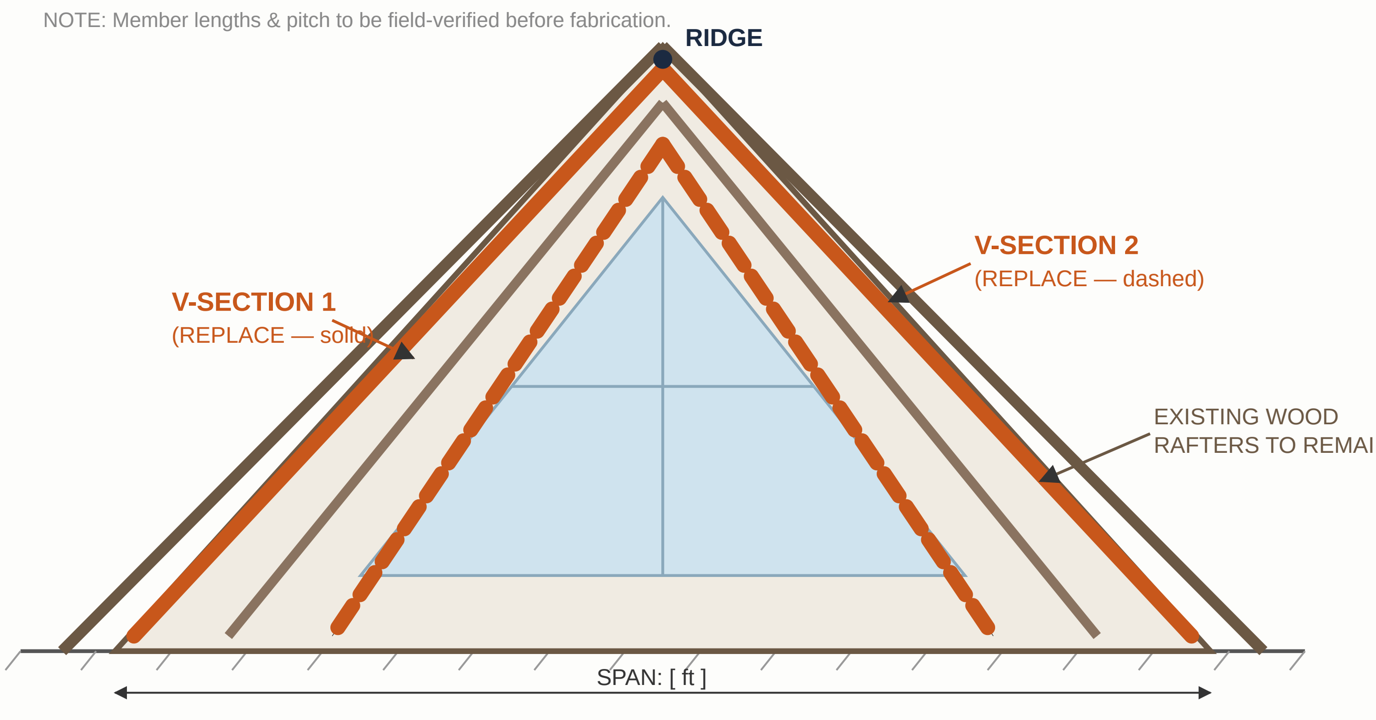
Same method, checked against residential floor loading (40 psf live + 10 psf dead) over the actual spans (13'-0" girders, 12'-0" joists) — mid-span deflection of the aluminum members is under 1/4", well inside the L/360 code limit:

Member	Weight (lb/ft)	Stiffness vs. wood	Bending capacity vs. wood
Existing girder pair — (2) wood 3"×11½" × 13'	9.0 ea	1.00×	1.00×
New girder pair — (2) alum. RT 3"×12"×3/16" 6061-T6 SELECTED	**6.5 ea**	**1.7×**	**4.3×**
Existing joist — wood 2"×11½" × 12' (7 pcs)	5.8	1.00×	1.00×
New joist — alum. RT 2"×12"×1/8" 6061-T6 (7 pcs) SELECTED	**4.1**	**1.5×**	**3.8×**

Purchase lengths: joists cut two per standard 24' bar (zero waste); girders supplied cut-to-length (13'-6" each) or from one 28'–30' bar, per EMS stock at time of order. If the 2"×12" joist profile is not in stock, alternate 2"×10"×3/16" with 2" seat shims performs equally (floor height allows, per owner). Member depth (11"–12") finalized with EMS availability — strength margins hold for both.

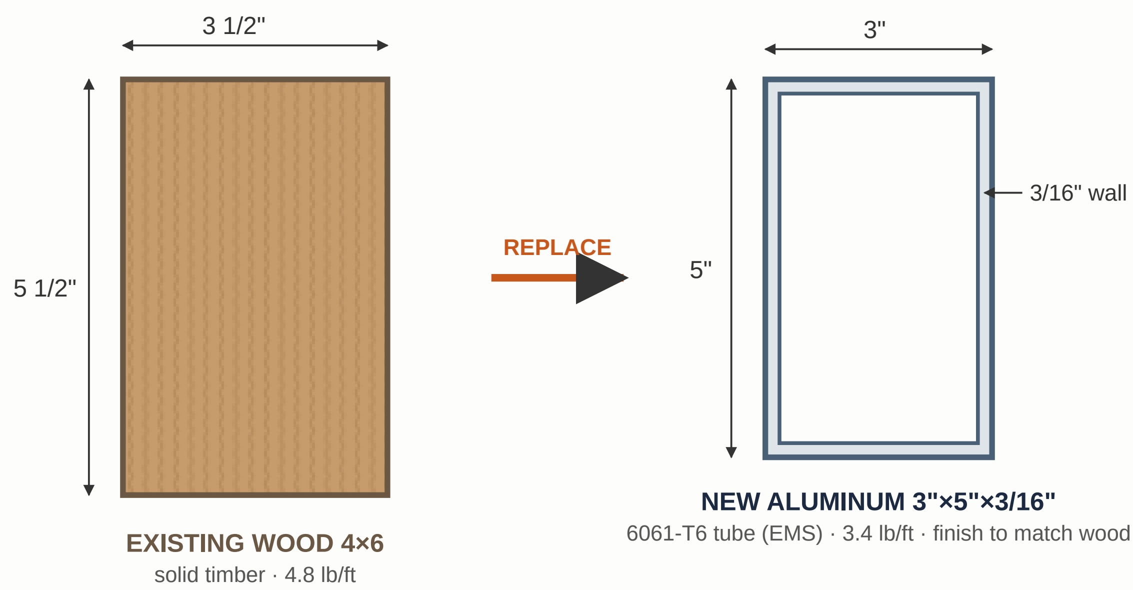
5. DRAWINGS

DWG-1 · GABLE ELEVATION — LOCATION OF V-SECTIONS TO BE REPLACED (N.T.S.)



Each "V-section" = one pair of sloped rafters meeting at the ridge. Exact members to be marked on site with owner prior to demolition.

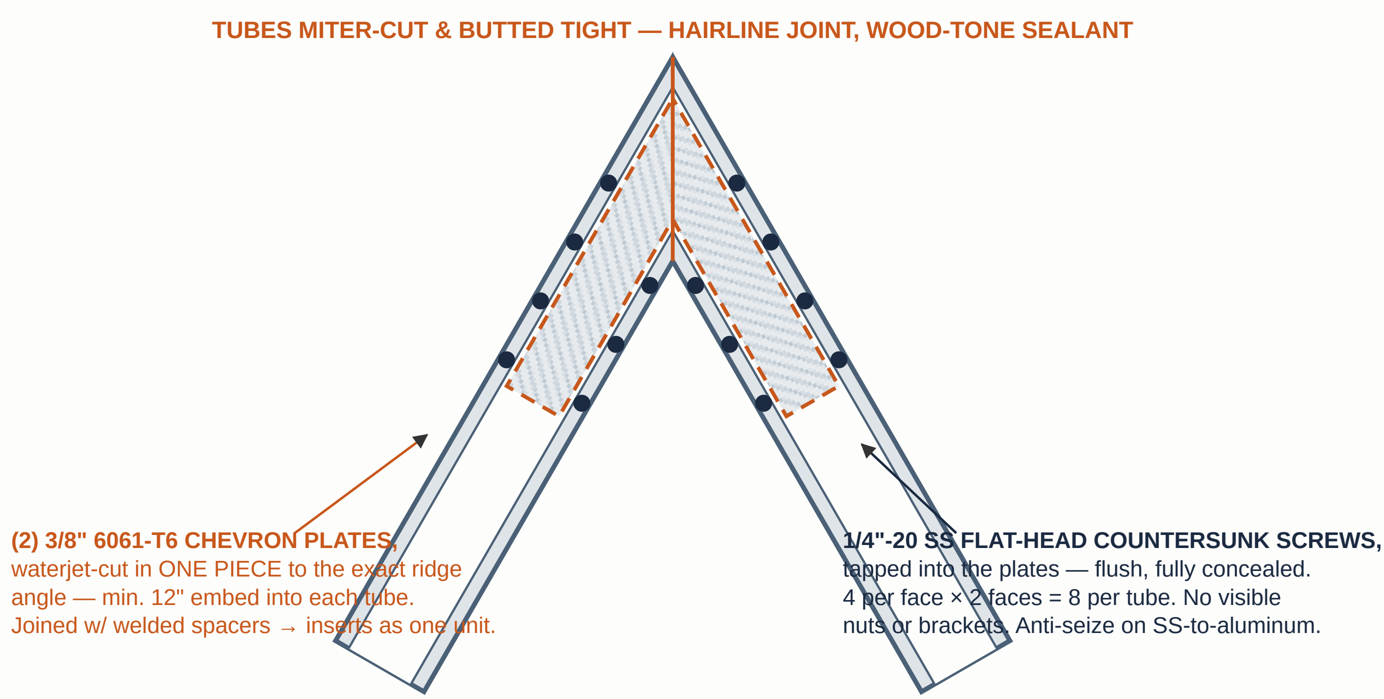
DWG-2 · CROSS-SECTION — EXISTING WOOD vs. NEW ALUMINUM (SCALE ~1:4)



Strength: 4.2× bending capacity, 1.4× stiffness of existing timber — 30% lighter.

Heavier alternate available: 4"×6"×3/16" 6061-T6 (6.7× wood bending strength, 2.7× stiffness) if a bulkier profile is preferred.

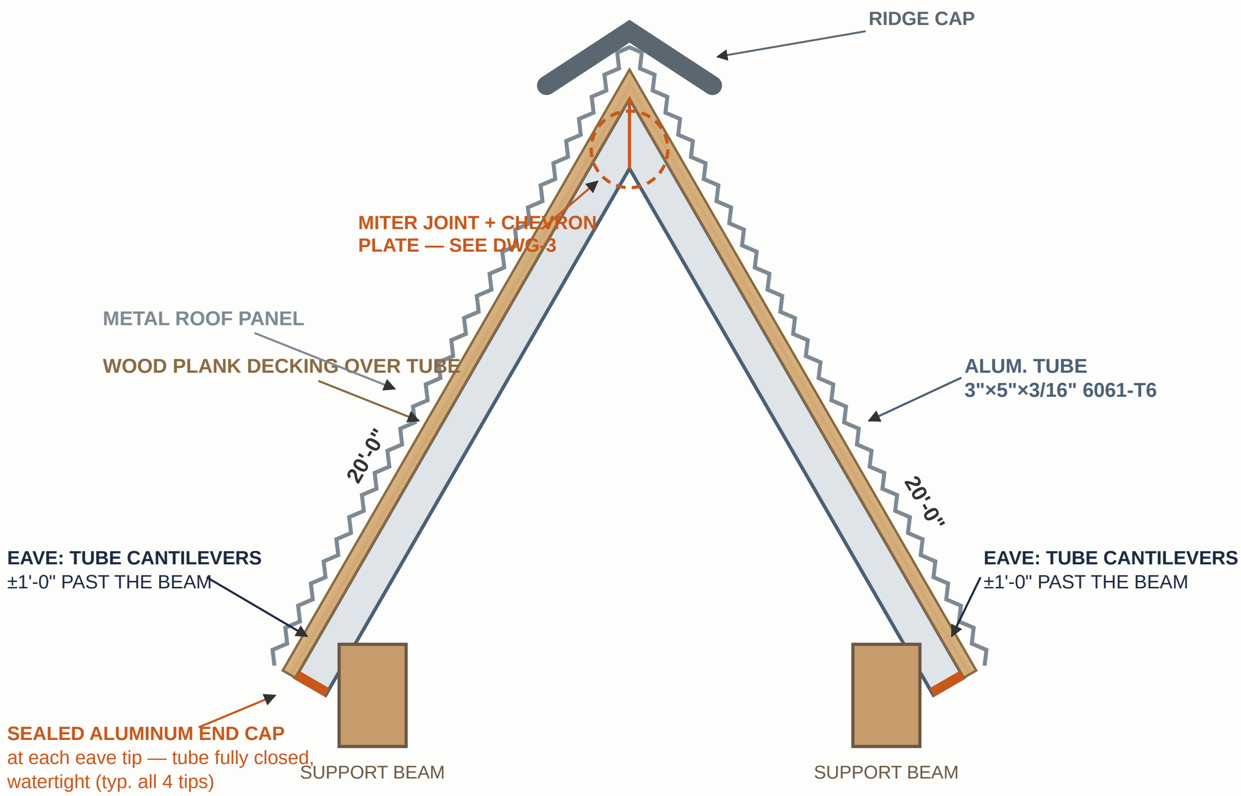
DWG-3 · RIDGE JOINT — MITER-CUT TUBES W/ CONCEALED ONE-PIECE CHEVRON PLATES (N.T.S.)



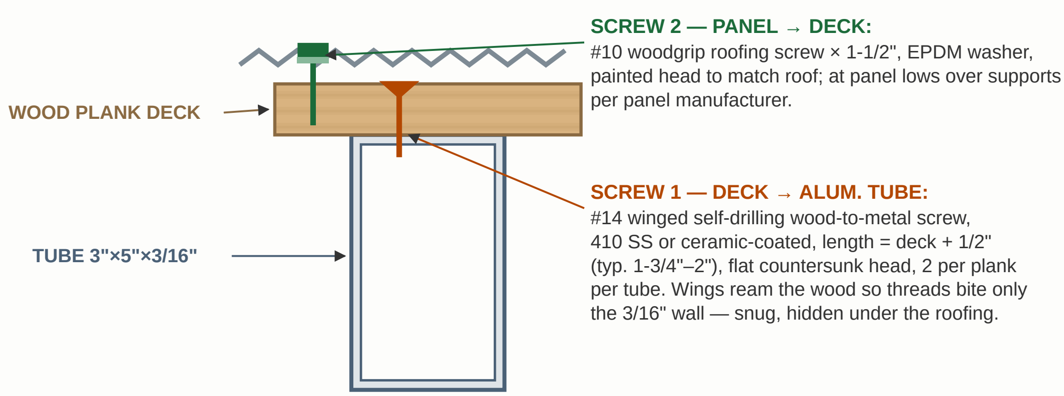
Joint capacity ≈ 8,000 lb per member (8 screws @ ~1,000 lb) — far above loads previously carried by the timber. 1/4" weep hole at each tube low point.

Bolted concealed joint keeps the 6061-T6 tubes at full strength (no weld heat-affected zone at the ridge). Field angle verified before plates are cut.

DWG-4 · GABLE FRONT VIEW — ROOF BUILD-UP, EAVE END CAPS & FASTENING (N.T.S.)



DETAIL B · DECK & PANEL FASTENING TO TUBE (section)



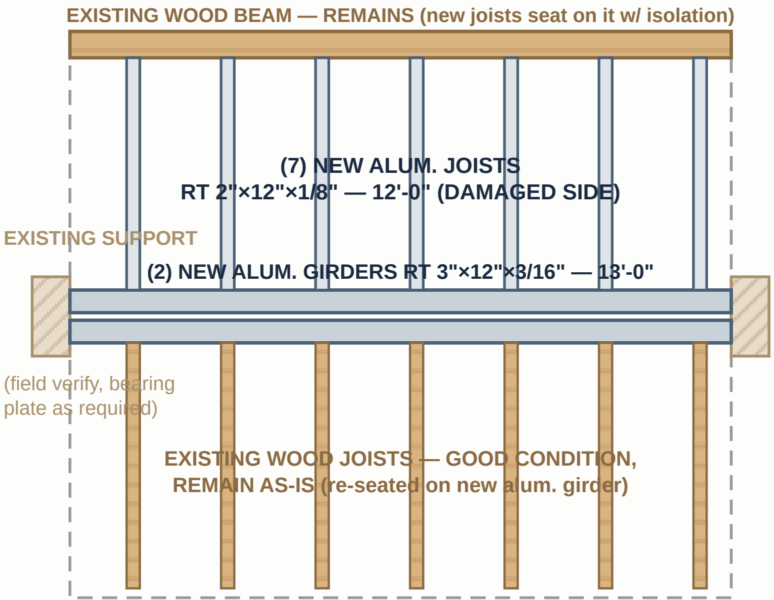
FASTENING NOTES:

1. No common zinc-plated screws against aluminum (galvanic corrosion) — 410 SS or ceramic-coated only; anti-seize on stainless.
2. Deck screws install from above and are concealed by the roofing panel — nothing visible from below.
3. Panel fastens to the wood deck (not directly to the tube) so roofing can be replaced later without touching the structure.
4. Drip edge at eave; sealed aluminum end caps at all four tube tips; 1/4" weep holes at tube low points.
5. Isolation barrier (butyl/bituminous tape) between pressure-treated wood and aluminum.

Deck planks and metal roof panel re-installed over the new aluminum members; all fasteners selected for aluminum compatibility.

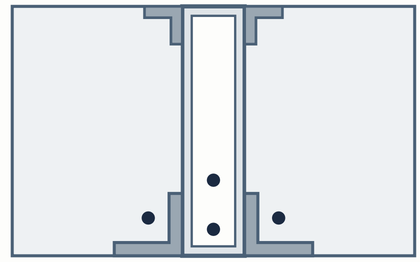
DWG-5 · FLOOR FRAMING — PLAN, JOIST SEAT & GIRDER-TO-COLUMN DETAILS (N.T.S.)

PLAN VIEW



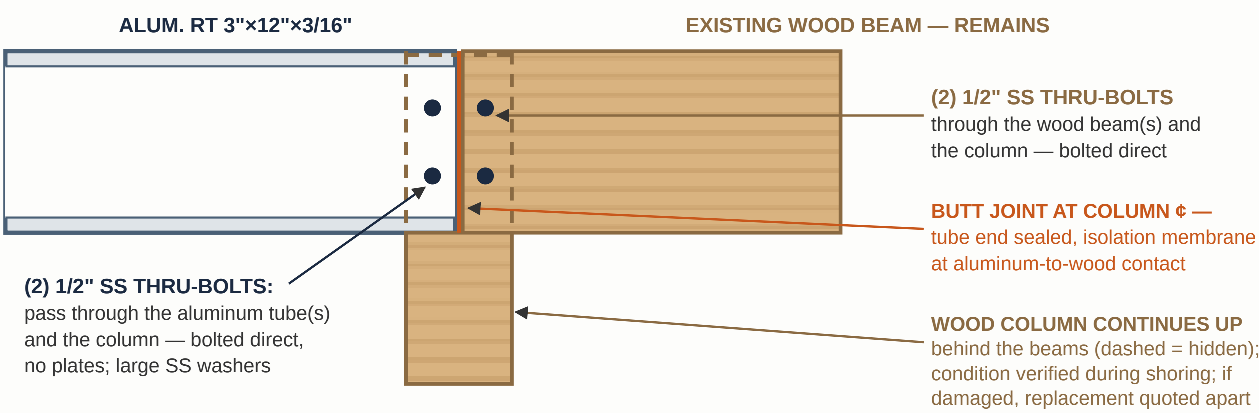
EACH NEW JOIST GOES IN THE EXACT LOCATION OF THE EXISTING ONE — SPACING & LAYOUT MATCH EXISTING

JOIST SEAT AT GIRDER (section)



3"×3"×1/4" ALUM. ANGLE SEATS
thru-bolted w/ SS hardware + top clip — replaces the corroded sheet-metal hangers. Nothing for termites to eat.

GIRDER CONNECTION AT COLUMN — BOLTED DIRECT TO THE COLUMN, NO PLATES (elevation)



Floor planks re-fastened to the new joists with #14 winged self-drilling screws (see DWG-4 Detail B — same fastener system as the roof).

6. PRICE — LUMP SUM

One single price covers the complete work described in this proposal. It includes:

PART A — ROOF V-SECTIONS

- Mobilization, scaffold/lift, protection & temporary shoring
- Careful demolition of (2) wood V-sections protecting the roof; haul & disposal
- (4) alum. RT 3"×5"×3/16" 6061-T6 × 20' (Eastern Metal Supply), chevron ridge inserts, sealed end caps, SS hardware
- Fabrication (miter joints, concealed inserts) & finish to match existing timber
- Installation, re-install wood deck & metal roof panel with correct fasteners, drip edge, ridge cap

PART B — FLOOR STRUCTURE

- Full temporary shoring of the floor during the entire replacement
- Demolition & disposal of (2) 3×12 girders + (7) 2×12 joists (damaged side only) and old hangers
- (2) alum. RT 3"×12"×3/16" girders + (7) RT 2"×12"×1/8" joists (EMS), angle seats, SS hardware
- Direct-bolted connections at columns and existing wood beam; isolation at all wood contact
- Every member set in the exact location of the existing one; floor planks re-screwed; cleanup

TOTAL — COMPLETE PROJECT (PARTS A & B), MATERIALS, LABOR & EQUIPMENT	$ [ ]

Lump-sum price. No itemized breakdown; scope is as specified in Sections 1–5. Any work outside this scope (e.g. replacement of a damaged column) is quoted separately by written change order before proceeding.

7. TERMS & CONDITIONS

- **Payment:** **[50%]** deposit to schedule and order material; balance due upon completion and owner walkthrough.
- **Schedule:** Approx. 2 weeks material & finish lead time. Site work (Parts A & B together): **10 to 14 working days with a 4-man crew** — demolition of one element proceeds while the next is staged and installed; weather permitting.
- **Field verification:** All lengths, pitches and connection geometry will be field-measured before fabrication.
- **Warranty:** One (1) year on workmanship. Material per manufacturer/supplier warranty.
- **Exclusions:** Building permit & engineering letter (available as add-on if required by the county), repair of concealed rot/damage in members not listed, roofing repairs beyond the work area, painting of existing wood, electrical work.
- Any additional work will be quoted and approved in writing before proceeding (change order).

JRG Welding Corp — Authorized Signature / Date

Client Acceptance — Signature / Date